

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

AMACO GmbH (a Coesia brand)

Ms. Dajana Smyrek

Vorarlberg

Austria

Subject: Application as Technical Project Engineer – Workflow & Vision Systems (m/f/d)

Dear Ms. Smyrek,

With a deep passion for optical precision and industrial automation, I am applying for the Technical Project Engineer position for Workflow and Vision Systems at AMACO GmbH. As a Photonics Engineer transitioning from my PhD, I offer extensive experience in developing, validating, and automating high-resolution vision and inspection systems that align perfectly with the requirements of Coesia's advanced packaging solutions.

During my previous role as a Machine Vision Engineer at INTECOL S.A.S., I successfully managed projects designing custom industrial vision inspection solutions using Halcon and C++. I was responsible for the end-to-end process: from evaluating the feasibility of lighting setups on the assembly line to ensuring strict tolerance compliance in automated sorting systems. This hands-on experience, combined with my ability to train operators and integrate complex systems, has given me a deep understanding of what it takes to commission robust industrial setups.

Currently, during my PhD at the Medical University of Innsbruck, I independently manage research projects that involve the end-to-end construction and validation of complex opto-mechatronic systems. I extensively use LabVIEW and Python to control metrology equipment and automate evaluation workflows, reinforcing my strong IT affinity and proactive analytical thinking. I am very comfortable acting as the technical interface between different disciplines, as I regularly coordinate with external manufacturers and engineering teams.

Originally from Colombia, I completed my Master's in Germany (Jena) and am currently rooted in the Alps during my PhD in Innsbruck (Tirol). I have excellent English skills and actively improve my German (B2 level). Given my proximity to Vorarlberg, I am highly interested in the region. Furthermore, my current Austrian Resident Permit as a researcher can be seamlessly converted into a regular working permit, allowing me to start immediately after May.

I am eager to bring my expertise in vision inspection systems and my structured project management approach to AMACO, and I look forward to an opportunity to interview with your team.

Sincerely,

Juan David Muñoz-Bolaños

Technical Project Engineer - Vision Systems

Medical University of Innsbruck

Juan David Muñoz-Bolaños

Technical Project Engineer - Vision Systems

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at | [LinkedIn Profile](#)



PROFESSIONAL SUMMARY

Physicist and Optical Engineer with comprehensive expertise in machine vision systems, optical metrology, and industrial automation. Proven track record in designing, qualifying, and automating integrated opto-mechatronic systems using LabVIEW, Python, and C++. Strong analytical approach to problem-solving with a customer-focused mindset and experience in managing end-to-end technical projects.

TECHNICAL COMPETENCIES

Vision Systems & Industrial Inspection

End-to-end development and laboratory testing of custom machine vision solutions for industrial assembly lines (Halcon, C++); expertise in advanced illumination concepts.

IT & Automation Integration

Proficient in LabVIEW, Python, and MATLAB for automating complex measurement workflows, data evaluation, and system control.

Project Management & Transfer

Independent coordination of complex technology projects from conception to validation; experience acting as a technical interface with external manufacturers and providing operator training.

Process Quality & Standards

Verification of optical tolerances (ZEMAX) and system capabilities (MSA/Gage R&R) under strict adherence to industrial standards.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher – Optical Development & Automation

Jan 2022 – Present

- **System Engineering:** End-to-end construction, integration, and validation of complex opto-mechatronic imaging systems.
- **Automation & Prototyping:** Extensive use of LabVIEW and Python to automate advanced measurement procedures and ensure process stability.
- **Project Interface:** Independent coordination of technical projects involving electronics specialists and continuous collaboration with external optics manufacturers.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

Research Assistant – Imaging & Optics

Jun 2020 – Dec 2021

- **Optical Engineering:** Conception and feasibility calculations to build a compact fiber-optic Raman spectrometer.
- **Data Analysis:** Applied machine learning and computational techniques to photonic data sets to evaluate and improve measurement workflows.

INTECOL S.A.S.

Medellín, Colombia

Junior Machine Vision System Engineer

Mar 2018 – Mar 2019

- **Industrial Machine Vision:** Development and validation of customized vision inspection systems for automated production lines (Halcon, C++).
- **Process Stability:** Evaluated the feasibility and measurement capabilities of various illumination configurations under tight tolerance guidelines directly on the conveyor belt.

EDUCATION

Medical University of Innsbruck

Ph.D. in Adaptive Optics and Microscopy

Austria

Jan 2022 – Present

Friedrich-Schiller-University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physical Engineering

Colombia

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** Selected for ASML and ZEISS Summer Schools; 2026 COLFUTURO Scholarship for top Colombian professionals.
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Highly competent),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Medical University of Innsbruck

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

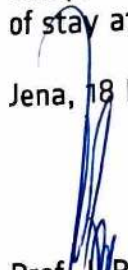
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

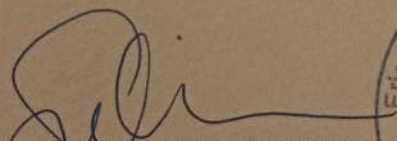
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stämmler

Dr. Thomas Stämmler
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passaporto № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

G. ANTIOQUIA

Forma dei tubuli / *Peperidera* e *Aggraderia*

Chas. Fawcett

[illegible]

